

Fourier transform activity

Student activity

Name _____ Date _____

- Construct graphs of data using the ***FourierTransform_ActivityGraph.ods*** file.
- Each graph should have data that appears random but has data points at one or more regular time intervals.
- Time axis from 0 s to 60 s
- Each graph contains two signals at the periods indicated below, with noise included in the signal. These are created in the *FourierTransform_ActivityGraph.ods* file.

1. (3, 7)

2. (4, 7)

3. (3, 8)

4. (4, 9)

5. (4, 6)

6. (3, 5)

7. (5, 9)

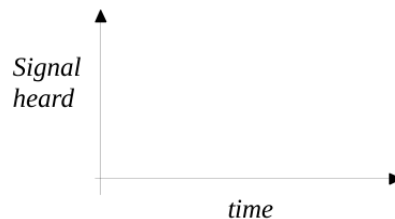
8. (5, 7)

9. (6, 9)

10. (5, 8)

50 s on time-axis = 7.50" $\rightarrow$ 1 s = 0.15"

- Intro
- Hand out a blank graph of signal vs. time to each student.



- Demonstrate GNU Radio sound of single, clean periodic pulses at 1.0 s intervals. Ask students to sketch what they think the pulse would look like on the graph.
- Have students discuss their graph with another student; then discuss as class:
 - Have a volunteer sketch their graph on the board
 - Why does your graph represent what you heard?
 - How is the *period* of a wave represented in your graph?
 - This is an example of a periodic signal.

- Signals that are not noise usually have some sort of pattern that is repetitious - periodic.
- Review the *period*, *amplitude*, and *frequency* of this signal, and how each is represented in a signal.
 - Then demonstrate GNU Radio pulses with noise added. Ask students to sketch what they think the signal would look like in this case.
 - Students compare with a neighbor.
 - Then discuss as class:
- What's different?
- What's the same?
- Noise is usually not periodic. Recall from yesterday: Signals carry information. When processing signals, it is necessary to distinguish the signal from the noise.
- Recall Tad's presentation with the photo filtering. This is filtering of amplitude.
- Another method scientists use is filtering by frequency. Scientists have methods to filter signals by frequency (or period).
 - Signal Processing Activity
 - Demonstrate process with the teacher graph
- explain masks.
- demonstrate sliding process to identify periods.
- explain that a periodic signal must fill at each period translation.
- explain how noise shows up – signals less than 1.0 (usually), and not periodic.
- Explain offset.
 - Group students into pairs.
 - Handout signal sheets and masks. Explain that they are to determine the periods in each signal graph; then replot the data showing only those periodic signals, without noise.
 - Have students sketch the Signal vs. Frequency graph that corresponds to their
 - Post Activity Discussion
 - What challenges are there determining the periodic signals in the total signal?
 - For the 3 s and 4 s signals, harmonics show up. How do you think we can distinguish whether the actual signal is the 3 s, 6 s, or 9 s?
 - Many signals are a mixing of multiple frequencies – next activity with Howard.
 - Scavenger hunt clue.
 - Motivation: Explain/remind students that waves are repetitious, e.g. water waves oscillate periodically. Information can be carried in the pattern, e.g. the morse code activity done previously.
 - Show sample graphs:
 - Example showing sum of sinusoidal signals. What periodic waves are in this?
 - Example with periodic pulse signal in noise.
- What repetitious (periodic) waves in this signal?
- Demonstrate masking method for determining the periodic waves in this.

- Lay the mask over the graph and slide the mask until each opening aligns with the data at one second intervals.
- By sliding the mask to match up with the data at each one second interval the period of the signal can be determined.
- Pair students up. Hand out graphs and masks to each group.
- Students are to use the mask method to determine the period of any waves in their graph.
- Have each group sketch what the graph would look like if there were no noise.
- Extension: On a signal vs. frequency graph, have each group plot signal vs. frequency.
- Scavenger hunt:
 - column (0 – 9) determined by amplitude of stronger signal
 - row (X – R) determined by frequency of weaker signal; use a table to convert frequency to a letter.